

PL — Polymers and Life

OCR B (Salters) H433 storyline **PL**. Polymers, proteins and DNA: condensation polymers, amines and amides, amino acid and protein chemistry, optical isomerism, enzyme catalysis, and structure determination from mass, IR and NMR spectra.

1. Amines and Amino Acids

Amines

Amine: An organic compound derived from ammonia (NH_3) by replacing one or more hydrogen atoms with alkyl or aryl groups. Primary amines have the formula RNH_2 , secondary amines R_2NH , and tertiary amines R_3N .

Preparation of amines:

- Reaction of a halogenoalkane with excess ammonia in a sealed tube at elevated temperature: $\text{RBr} + 2\text{NH}_3 \rightarrow \text{RNH}_2 + \text{NH}_4\text{Br}$. Excess ammonia is used to minimise further substitution.
- Reduction of a nitrile with LiAlH_4 in dry ether or by catalytic hydrogenation: $\text{RCN} + 4[\text{H}] \rightarrow \text{RCH}_2\text{NH}_2$.
- Reduction of a nitro compound (important for aromatic amines): $\text{C}_6\text{H}_5\text{NO}_2 + 6[\text{H}] \rightarrow (\text{Sn/conc. HCl, reflux}) \text{C}_6\text{H}_5\text{NH}_2 + 2\text{H}_2\text{O}$.

Base properties of amines: Amines act as Brønsted-Lowry bases because the lone pair on nitrogen can accept a proton: $\text{RNH}_2 + \text{H}^+ \rightarrow \text{RNH}_3^+$. Aliphatic amines are stronger bases than ammonia because the electron-donating alkyl groups increase electron density on the nitrogen, making the lone pair more available for protonation. Aromatic amines (e.g. phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$) are weaker bases than ammonia because the nitrogen lone pair is delocalised into the benzene ring, reducing its availability.

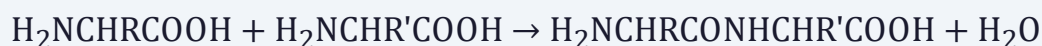
Amino Acids

Amino acid: An organic molecule containing both an amine group ($-\text{NH}_2$) and a carboxylic acid group ($-\text{COOH}$). The α -amino acids found in proteins have both groups attached to the same carbon atom (the α -carbon), along with a hydrogen atom and a variable side chain (R group).

Zwitterions: In solution at the isoelectric point, amino acids exist predominantly as zwitterions — the amine group is protonated ($-\text{NH}_3^+$) and the carboxylic acid group is deprotonated ($-\text{COO}^-$). This gives the molecule both a positive and a negative charge but an overall neutral charge. Zwitterions have high melting points and are soluble in water due to their ionic character.

In acidic solution, the COO^- group is protonated, giving a positive ion. In basic solution, the NH_3^+ group is deprotonated, giving a negative ion.

Peptide Bonds and Proteins



A peptide bond ($-\text{CONH}-$) forms by a condensation reaction between the $-\text{COOH}$ of one amino acid and the $-\text{NH}_2$ of another, releasing water. Proteins are polypeptides formed from many amino acids linked by peptide bonds. Peptide bonds can be broken by hydrolysis (acid or base conditions, or enzymatically).

Levels of Protein Structure

A protein is not a floppy string. The chain folds into one particular shape, and that shape is what does the job — catalysing a reaction, carrying oxygen, forming a fibre. Chemists describe the folding at three levels, and the useful question at each level is *which bond holds it together*.

Primary structure: The sequence of amino acid residues in the polypeptide chain, held together by covalent **peptide bonds**.

Secondary structure: The regular local folding of the chain — the **α -helix** and the **β -pleated sheet** — held by **hydrogen bonds between the $\text{C}=\text{O}$ of one peptide link and the $\text{N}-\text{H}$ of another** along the backbone.

Tertiary structure: The overall three-dimensional shape of the whole folded chain, held by

interactions between the **R groups**: hydrogen bonds, ionic attractions, sulfur–sulfur (disulfide) bridges and instantaneous dipole–induced dipole bonds.

Primary structure is simply the order of the residues, conventionally written starting from the end that still carries a free -NH_2 group. It is set by the base sequence of the gene (section 4), and because peptide bonds are covalent it survives everything short of hydrolysis. Everything that follows is a consequence of it: the same sequence, folding by the same rules, always arrives at the same shape.

Secondary structure comes from the peptide link itself. Every link contributes a C=O , polar with δ^- on the oxygen, and an N-H , polar with δ^+ on the hydrogen. Those two groups repeat all the way along the backbone, so the chain can hydrogen bond to itself:

- **α -helix**: the chain coils into a spiral, and each C=O hydrogen bonds to the N-H four residues further along the *same* chain. The hydrogen bonds run roughly parallel to the axis of the coil, and the R groups point outwards away from it.
- **β -pleated sheet**: sections of chain lie side by side, and each C=O hydrogen bonds to an N-H on the neighbouring section. Here the hydrogen bonds run across between the strands, and the sheet is slightly corrugated — hence the name "pleated".

Notice what the R groups are *not* doing at this level. Secondary structure is a backbone effect, and the hydrogen bonding is the same chemistry whichever amino acids happen to be present.

Tertiary structure is where the side chains take over. The helices and sheets fold back on one another and the chain is locked in place by four kinds of interaction between R groups:

Interaction	Between which R groups	Relative strength
Sulfur–sulfur (disulfide) bridge, -S-S-	Two cysteine residues, whose $\text{-CH}_2\text{SH}$ groups are oxidised together	Strongest — a covalent bond
Ionic attraction (a "salt bridge")	A protonated basic side chain (-NH_3^+ , e.g. lysine) and a deprotonated acidic one (-COO^- , e.g. glutamic acid)	Strong
Hydrogen bond	Polar side chains carrying -OH , -NH_2 , -COOH or -CONH_2	Moderate

Instantaneous dipole–induced dipole bond	Non-polar hydrocarbon side chains, such as those of valine and leucine	Weakest individually, but there are very many of them
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Why the Structure Decides the Properties

Solubility. In water a globular protein folds so that the polar and charged R groups finish on the outside, where they can hydrogen bond to water or attract ions in solution, while the non-polar hydrocarbon R groups are packed into the interior and held there by instantaneous dipole–induced dipole bonds. The outside of the molecule therefore looks polar to water, and the protein dissolves. Fibrous proteins such as keratin do the opposite: long chains lie alongside one another, cross-linked by many hydrogen bonds and disulfide bridges, which leaves little for water to interact with, so the protein is insoluble, strong and useful as a structural material.

Denaturing. Heating a protein gives the chain enough energy to break the hydrogen bonds, ionic attractions and instantaneous dipole–induced dipole bonds that hold the tertiary and secondary structure. The chain unfolds, the shape is lost, and with it the function: an enzyme's active site no longer fits its substrate (section 5), and a soluble protein such as egg albumin exposes its non-polar interior and coagulates instead of dissolving. Extremes of pH do the same job by a different route — added H^+ protonates COO^- side chains and added OH^- deprotonates NH_3^+ side chains, so the ionic attractions simply stop existing.

Two things survive. The peptide bonds are covalent, so the primary structure is untouched — denaturing is not hydrolysis, and the same residues are still there in the same order. Disulfide bridges are covalent as well, which is why breaking them needs a reducing agent rather than heat, and why hair, cross-linked by disulfide bridges in keratin, has to be chemically reduced and then re-oxidised in a new shape to be permanently curled or straightened.

Exam Tip: Marks here are earned by naming the interaction *and* the level it holds. "Hydrogen bonding" on its own is ambiguous: hydrogen bonds between backbone C=O and N–H groups hold the secondary structure, while hydrogen bonds between R groups hold the tertiary structure. Say which. And never offer the peptide link as an intermolecular bond — it is covalent and it belongs to the primary structure.

2. Optical Isomerism

Look again at the general structure of an α -amino acid. The α -carbon carries four groups: $-\text{NH}_2$, $-\text{COOH}$, $-\text{H}$ and the side chain R. For every amino acid except glycine those four groups are all different, and that single fact gives rise to a second kind of stereoisomerism — one that governs how proteins fold, how enzymes select their substrates, and why a drug can be a cure in one form and a poison in the other.

Chiral centre: A carbon atom bonded to four *different* atoms or groups. A molecule containing one is described as **chiral**.

Enantiomers (optical isomers): The two forms of a chiral molecule, which are non-superimposable mirror images of one another.

"Non-superimposable" is the key phrase and it is worth getting a feel for physically. Your two hands are mirror images, but no matter how you rotate them you cannot lay one on the other so that thumb covers thumb and palm covers palm. A chiral molecule is the same: the mirror image is a genuinely different molecule, not the same molecule viewed from another angle.

Finding the Chiral Centre

To test a carbon atom, list the four things attached to it and ask whether any two are identical. If all four differ, that carbon is a chiral centre and is usually marked with an asterisk.

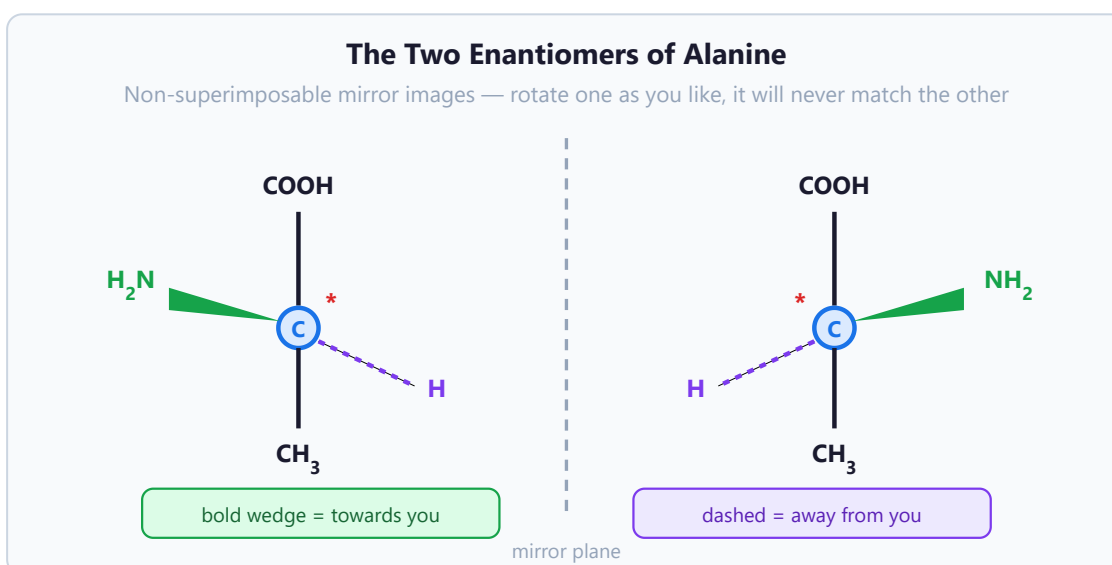
Molecule	Groups on the carbon in question	Chiral?
Glycine, $\text{H}_2\text{NCH}_2\text{COOH}$	$-\text{NH}_2$, $-\text{COOH}$, $-\text{H}$, $-\text{H}$	No — two hydrogens are the same
Alanine, $\text{CH}_3\text{CH}(\text{NH}_2)\text{COOH}$	$-\text{NH}_2$, $-\text{COOH}$, $-\text{H}$, $-\text{CH}_3$	Yes
Butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5$	$-\text{OH}$, $-\text{H}$, $-\text{CH}_3$, $-\text{C}_2\text{H}_5$	Yes
Propan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	$-\text{OH}$, $-\text{H}$, $-\text{CH}_3$, $-\text{CH}_3$	No — two methyl groups are the same

2-hydroxypropanoic acid (lactic acid)	$-\text{OH}$, $-\text{H}$, $-\text{CH}_3$, $-\text{COOH}$	Yes
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Glycine is therefore the only one of the twenty amino acids coded for in proteins that is *not* chiral. All the others exist as a pair of enantiomers, and in living organisms essentially only one of each pair — the L-form — is used to build proteins.

Drawing Optical Isomers

The marks in this question are for showing three dimensions properly, so use the wedge-and-dash convention: two ordinary lines in the plane of the paper, one bold wedge coming towards you, one dashed line going away. Draw the first isomer, then draw a vertical mirror line and reflect every group across it.



Why the Two Forms Are Not the Same

A pair of enantiomers have identical melting point, boiling point, density, solubility and ordinary chemical reactivity, because every bond in one is the same as the corresponding bond in the other. They differ in only two respects, and both follow from their handedness:

- They rotate the plane of **plane-polarised light** by equal amounts in opposite directions — one clockwise, one anticlockwise. This is what "optical" isomerism is named after, and it is how the two forms are told apart in the laboratory, using a polarimeter.
- They behave differently towards *other* chiral molecules. Enzymes, receptors and antibodies are built from L-amino acids and are themselves chiral, so a chiral active site fits one enantiomer of a substrate or drug and not the other — exactly the lock-and-key idea that section 5 returns to.

Racemic mixture (racemate): An equal mixture of the two enantiomers. It shows no rotation of plane-polarised light, because the two opposite rotations cancel exactly. Ordinary laboratory synthesis from achiral starting materials always gives a racemate, since the reagent is equally likely to attack either face of a planar intermediate.

Why This Matters for Medicines

Because a receptor site can distinguish enantiomers, the two forms of a chiral drug can have quite different effects. Only one enantiomer of thalidomide acts as the intended sedative; the other is a teratogen, and the two interconvert in the body, so even administering the pure "safe" form would not have prevented harm. In the case of ibuprofen only the S-enantiomer is the active anti-inflammatory, but the body converts much of the inactive R-form into it, so the racemate can still be sold. Producing a single enantiomer is expensive — it needs a chiral starting material, a chiral catalyst, an enzyme, or separation of the racemate — but it halves the dose needed and removes whatever the unwanted enantiomer would have done.

Exam Tip: Three marks are regularly lost here. First, when asked to identify chiral centres in a large molecule, check *every* carbon systematically rather than guessing — and remember a carbon in a C=C or a benzene ring can never be chiral, because it has only three groups attached. Second, when asked to draw the isomers, you must use wedges and dashes; two flat structures drawn side by side score nothing. Third, if asked how the isomers differ, "they rotate plane-polarised light in opposite directions" is the physical difference — do not say they have different melting points or different chemical reactions with ordinary achiral reagents, because they do not.

3. Polymers

Addition Polymers

Formed from alkene monomers by breaking C=C double bonds. No by-product is formed. Examples: poly(ethene), poly(propene), PVC, PTFE. They are non-biodegradable and chemically inert (strong C–C backbone).

Condensation Polymers

Condensation polymerisation: Monomers join together with the loss of a small molecule (usually water) at each linkage. The monomers must have two functional groups each (they are bifunctional).

Polyesters: Formed from a diol ($\text{HO}-\text{R}-\text{OH}$) and a dicarboxylic acid ($\text{HOOC}-\text{R}'-\text{COOH}$), or from a hydroxycarboxylic acid ($\text{HO}-\text{R}-\text{COOH}$). The linkage is an ester bond ($-\text{COO}-$). Example: Terylene (PET) from ethane-1,2-diol and benzene-1,4-dicarboxylic acid.

Polyamides: Formed from a diamine ($\text{H}_2\text{N}-\text{R}-\text{NH}_2$) and a dicarboxylic acid ($\text{HOOC}-\text{R}'-\text{COOH}$), or from an amino acid. The linkage is a peptide/amide bond ($-\text{CONH}-$). Examples: nylon-6,6 from 1,6-diaminohexane and hexanedioic acid; Kevlar from 1,4-diaminobenzene and benzene-1,4-dicarboxylic acid.

Biodegradability: Condensation polymers are generally biodegradable because the ester or amide bonds can be hydrolysed by water, acids, bases, or enzymes. Addition polymers are generally non-biodegradable. Biodegradable polymers reduce landfill waste but may produce greenhouse gases (CO_2 , CH_4) during decomposition.

Exam Tip: When drawing condensation polymer repeat units, remember to show the linkage (ester or amide bond) and indicate where water was lost. The repeat unit should be enclosed in brackets with continuation bonds on each side. Be able to identify the monomers from a given polymer structure by breaking the ester or amide bonds and adding back H and OH.

Naming Nylons

Nylon is a family of polyamides rather than a single substance, and the numbers in the name say exactly which monomers were used. **The digits count carbon atoms: the first is the number of carbons in the diamine, the second the number in the dicarboxylic acid** — counting the two carboxyl carbons themselves. A nylon written with only one digit was made from a single monomer carrying both functional groups, and the digit is the number of carbons in that monomer.

Polymer	Monomer(s)	Repeat unit
Nylon-6,6	1,6-diaminohexane, $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (6 C), with hexanedioic acid, $\text{HOOC}(\text{CH}_2)_4\text{COOH}$ (6 C)	$-\text{[NH}(\text{CH}_2)_6\text{NHCO}(\text{CH}_2)_4\text{CO]}$ —

Nylon-6,10	1,6-diaminohexane, $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (6 C), with decanedioic acid, $\text{HOOC}(\text{CH}_2)_8\text{COOH}$ (10 C)	$-\text{[NH}(\text{CH}_2)_6\text{NHCO}(\text{CH}_2)_8\text{CO]}-$
Nylon-6	One monomer only: 6-aminohexanoic acid, $\text{H}_2\text{N}(\text{CH}_2)_5\text{COOH}$ (6 C), or its cyclic amide caprolactam	$-\text{[NH}(\text{CH}_2)_5\text{CO]}-$

The counting trap is in the acid. Hexanedioic acid has six carbons, but two of them are the carbonyl carbons of the two $-\text{COOH}$ groups, so only *four* CH_2 groups appear between the $\text{C}=\text{O}$ groups in the repeat unit. The diamine is not like that: all six of its carbons survive as the $(\text{CH}_2)_6$ chain, because its functional groups are nitrogen atoms on the ends rather than carbon atoms. So nylon-6,6 shows $(\text{CH}_2)_6$ and $(\text{CH}_2)_4$, and nylon-6,10 shows $(\text{CH}_2)_6$ and $(\text{CH}_2)_8$.

Nylon-6 is the odd one out because it needs no partner. 6-aminohexanoic acid already carries an $-\text{NH}_2$ at one end and a $-\text{COOH}$ at the other, so it condenses with itself, losing one water molecule per link, in exactly the way an amino acid does when it builds a protein. Industrially the polymer is usually made instead from **caprolactam**, the cyclic amide of that same acid, whose ring is opened and joined up without any small molecule being lost. Either route gives the same $-\text{[NH}(\text{CH}_2)_5\text{CO]}-$ repeat unit.

The numbers matter for the properties as well as for the name. Every amide link along a chain can hydrogen bond to an amide link on a neighbouring chain, and it is those hydrogen bonds that make nylon strong. Nylon-6,10 spaces its amide links further apart than nylon-6,6 does, because of the longer hydrocarbon section contributed by the C_{10} acid. Fewer hydrogen bonds per unit length means a lower melting point, a more flexible polymer and less water taken up — which is why nylon-6,10 is chosen for brush bristles and monofilament, and nylon-6,6 where stiffness and heat resistance matter.

Exam Tip: To name a nylon from its structure, find the nitrogen atoms and count the carbons between them — that gives the first digit. Then count from one $\text{C}=\text{O}$ carbon to the next, *including* both carbonyl carbons — that gives the second. If the repeat unit contains only one nitrogen and only one $\text{C}=\text{O}$ it came from a single monomer, so it takes a single number: count every carbon in the repeat unit.

4. DNA and RNA

Proteins are condensation polymers of amino acids. DNA and RNA are condensation polymers too, built from a different monomer: the nucleotide.

Nucleotide: A monomer with three components joined together — a **phosphate** group, a **sugar** (deoxyribose in DNA, ribose in RNA) and a nitrogen-containing **base**. The structures of all three are given on the H433 Data Sheet, so you do not need to memorise them; you do need to be able to identify which part is which and where the condensation reactions occur.

Building the Polymer

Two different condensation reactions hold a nucleic acid together, and water is lost in each:

- **Base to sugar:** an –NH group of the base condenses with an –OH group on carbon 1 of the sugar, so each sugar carries a base.
- **Phosphate to sugar:** an –OH of the phosphate group condenses with an –OH on the sugar, forming a phosphate ester link. Repeating this joins sugar–phosphate–sugar–phosphate into the **sugar–phosphate backbone**, with the bases sticking out sideways.

Because the links are made by condensation, they can be broken by **hydrolysis** — the same relationship as for polyesters and polyamides earlier in this storyline.

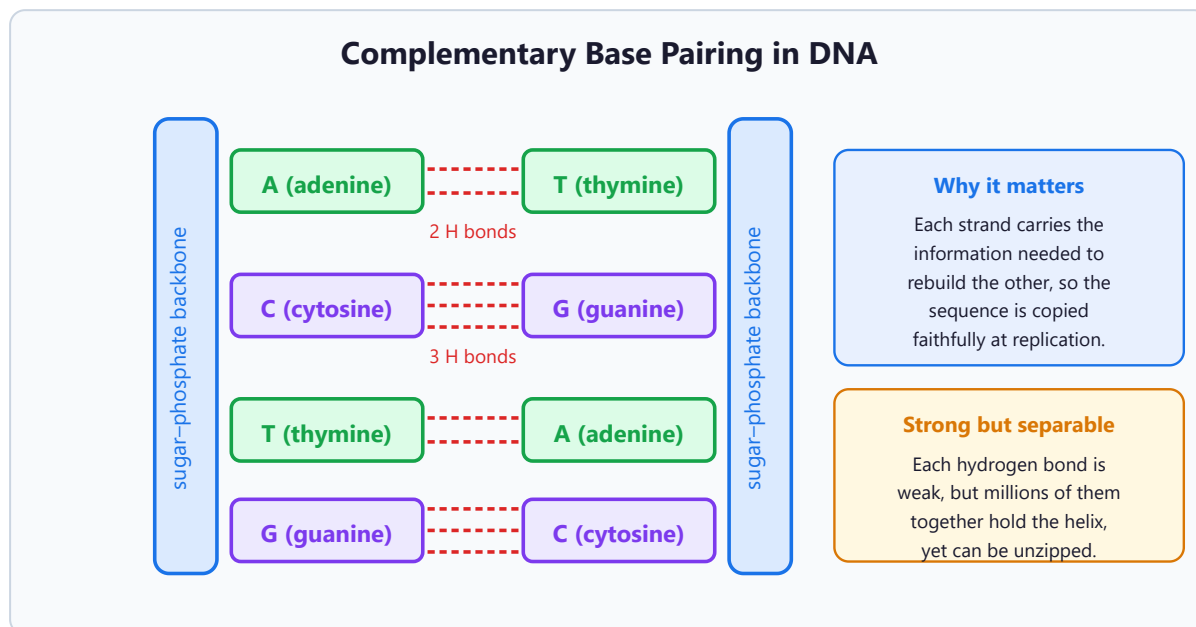
The Bases and Base Pairing

Polymer	Sugar	Bases	Structure
DNA	Deoxyribose	Adenine (A), Thymine (T), Cytosine (C), Guanine (G)	Two strands wound into a double helix
RNA	Ribose	Adenine (A), Uracil (U) , Cytosine (C), Guanine (G)	Normally a single strand

The two DNA strands are held together by **hydrogen bonds** between the bases, and the pairing is strictly complementary:

A ... T (two hydrogen bonds) C ... G (three hydrogen bonds)

A pairs with T (or with U in RNA) and C pairs with G because those are the only combinations in which an N–H or O–H hydrogen-bond donor on one base lines up with a lone pair on N or O on the other, at the right distance and orientation, for every hydrogen bond in the pair. A purine always pairs with a pyrimidine, which keeps the two strands a constant distance apart all the way along.



Why Hydrogen Bonding is the Right Strength

An individual hydrogen bond is weak compared with a covalent bond, but there are one for every base pair along the whole molecule, so collectively they hold the double helix firmly. That combination is exactly what replication needs: the two strands can be unzipped by an enzyme without breaking any covalent bond, and each separated strand then acts as a template on which the complementary bases assemble in only one possible order. The sequence is therefore copied faithfully.

From DNA to Protein

The base sequence of DNA is a code. Read in groups of three — a **triplet** or **codon** — it specifies the order in which amino acids are joined:

1. A section of DNA is used as the template to build a strand of messenger RNA, with U appearing wherever the DNA had A.
2. The mRNA is read three bases at a time, and each codon specifies one amino acid. The Data Sheet gives a short list of codons, for example GGU for glycine, GCC for alanine and GUC for valine.
3. The amino acids are joined in that order by peptide links, giving a protein with a defined primary structure.

Because the primary structure determines how the chain folds, the base sequence ultimately determines the protein's three-dimensional shape and therefore what it does.

Worked Example — Using the Genetic Code:

A section of an mRNA strand reads GGU GCC GUC. Using the Data Sheet codons (glycine GGU, alanine GCC, valine GUC), give the amino acid sequence and the DNA template from which it was transcribed.

Amino acid sequence: **glycine – alanine – valine**

The DNA template is complementary to the mRNA, with T replacing U and each base paired A–U/T and C–G:

mRNA	G	G	U	G	C	C	G	U	C
DNA	C	C	A	C	G	G	C	A	G

The **Human Genome Project** set out to read the entire base sequence of human DNA. A working draft was announced in 2001, the finished sequence followed in 2003, and the last remaining gaps — in highly repetitive regions that the original methods could not resolve — were closed by the Telomere-to-Telomere consortium, which published a gapless sequence in 2022 and completed the Y chromosome in 2023. The sequence underpins work on inherited disease and on medicines matched to a patient's own genetics.

Exam Tip: Two marks are routinely dropped here. First, DNA and RNA are *condensation* polymers — say that water is lost at each link. Second, when asked why base pairing is specific, do not just say "A pairs with T": explain that the hydrogen-bond donors and acceptors on the two bases match up in size, position and orientation, and give the 2 and 3 hydrogen bonds for A–T and C–G.

5. Enzyme Catalysis and Molecular Recognition

Enzyme: A protein that acts as a catalyst for a biological reaction. The substrate binds to a small region called the **active site**, whose shape and chemistry are created by the way the protein chain folds — that is, the active site is part of the tertiary structure.

Enzymes are extraordinarily effective catalysts, speeding reactions by factors of millions while working at body temperature and near-neutral pH. They achieve this the same way any catalyst does: by providing an alternative route of lower activation enthalpy.

The Characteristics of Enzyme Catalysis

Characteristic	Observation	Explanation in terms of the active site
Specificity	An enzyme catalyses one reaction, or one narrow class of reaction, and ignores other molecules	Only a substrate with the right size, shape, charge distribution and hydrogen-bonding groups fits into the active site and binds
Temperature sensitivity	Rate rises with temperature to an optimum, then falls sharply — unlike an ordinary catalysed reaction	Above the optimum, the intermolecular bonds holding the tertiary structure (hydrogen bonds, ionic bonds, sulfur–sulfur bridges, instantaneous dipole–induced dipole bonds) are broken. The chain unfolds, the active site loses its shape, and the enzyme is denatured
pH sensitivity	Rate peaks at an optimum pH and falls either side	Changing pH protonates or deprotonates -NH_2 and -COOH side chains, destroying the ionic bonds that hold the tertiary structure and altering the charges inside the active site
Competitive inhibition	A similar-shaped molecule slows the reaction, but the effect can be overcome by adding more substrate	The inhibitor is close enough in shape to occupy the active site, blocking the substrate. Substrate and inhibitor compete for the same sites, so raising the substrate concentration restores the rate

The Rate Curve

Plotting rate against substrate concentration for an enzyme-catalysed reaction gives a curve with a distinctive shape: it rises steeply at first, then levels off to a horizontal plateau.

Region	Order with respect to substrate	Why
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Low [substrate]	First order — rate is proportional to [substrate]	Most active sites are empty, so doubling the substrate concentration doubles the number of successful encounters
High [substrate]	Zero order — rate is independent of [substrate]	Every active site is already occupied, so the enzyme is saturated. Adding more substrate cannot speed things up; the rate now depends only on how fast the enzyme can turn substrate into product

The levelling-off is strong evidence for the active-site model itself: a simple collision between two free molecules would not saturate.

Molecular Recognition and Medicines

The same idea — a molecule being recognised by a binding site because it fits in three dimensions — explains how most medicines work.

Pharmacophore: The part of a drug molecule responsible for its biological activity, the group of atoms that actually interacts with the receptor site. Other groups on the molecule **modify** it — altering how strongly it binds, how soluble it is, how long it survives in the body, or how selective it is.

Binding to a receptor site depends on three-dimensional fit, and the interactions involved are the familiar ones: hydrogen bonds, ionic bonds between charged groups, instantaneous dipole-induced dipole bonds, and sometimes covalent bond formation. For binding to happen the molecule must match the site in **size**, in **shape**, in the **orientation** of its functional groups, and in its ability to **form bonds** at the right points. This is why two optical isomers of the same drug, identical in every other respect, can differ completely in activity: only one of them fits.

Many medicines are deliberately designed as **enzyme inhibitors**. A molecule shaped like the natural substrate occupies the active site, the enzyme cannot do its job, and a biochemical pathway is shut down. Methanol poisoning is treated on the same principle in reverse: methanol is only dangerous because an enzyme oxidises it to methanoic acid, so the patient is given ethanol, which competes for the same active site and is oxidised instead, allowing the methanol to be excreted unchanged.

Exam Tip: Denaturing is not the same as inhibiting. Denaturing destroys the tertiary structure by breaking intermolecular bonds, and is usually irreversible; competitive inhibition leaves the enzyme

intact and can be reversed by adding more substrate. If a question tells you the effect is overcome by increasing substrate concentration, that is your signal for competitive inhibition — say so explicitly.

6. High-Resolution Mass Spectrometry

Mass spectrometry was introduced in **WM**: the molecular ion peak M^+ gives the relative molecular mass, and the fragments below it show how the molecule breaks up. An ordinary low-resolution instrument reports m/z as a whole number, and that is a real limitation, because a great many different formulae share the same whole number. A **high-resolution** instrument measures m/z to four or five decimal places, and that is enough to fix the molecular formula at one answer.

Nominal mass: the mass of an ion worked out from whole-number masses ($H = 1$, $C = 12$, $N = 14$, $O = 16$). **Precise relative isotopic mass:** the mass of one specific isotope measured against $^{12}\text{C} = 12.000000$ exactly, quoted to several decimal places. A high-resolution mass spectrum gives the precise mass of the molecular ion, and only one molecular formula fits it.

Only ^{12}C has a whole-number mass, and that is because the whole scale is *defined* from it: one unit is one twelfth of the mass of a ^{12}C atom. Every other isotope comes out slightly off a whole number, because nuclei differ in how tightly their nucleons are bound and so in how much mass is lost as binding energy. A nucleus more tightly bound per nucleon than ^{12}C comes out **below** its nominal mass; one less tightly bound comes out **above** it.

Isotope	Precise relative isotopic mass	Nominal mass	Above or below?
^1H	1.007825	1	Above
^{12}C	12.000000 (exact, by definition)	12	—
^{14}N	14.003074	14	Above
^{16}O	15.994915	16	Below
^{32}S	31.972071	32	Below

^{35}Cl	34.968853	35	Below
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The consequence is that hydrogen and oxygen pull a formula's precise mass in opposite directions, so the errors do not cancel. Two formulae with the same nominal mass end up measurably apart. The standard demonstration is the set of ions at nominal m/z 28, which a low-resolution instrument cannot tell apart at all:

Ion	Working	Precise m/z
CO^+	$12.000000 + 15.994915$	27.9949
N_2^+	2×14.003074	28.0061
CH_2N^+	$12.000000 + 2.015650 + 14.003074$	28.0187
C_2H_4^+	$24.000000 + 4.031300$	28.0313

The four are spread over 0.036 of a mass unit. An instrument that reads to four decimal places separates them without difficulty; one that reads to the nearest whole number sees a single peak at 28.

Worked example — deducing a molecular formula. The high-resolution mass spectrum of a compound shows M^+ at m/z 88.0524. Four formulae have a nominal mass of 88. Which one is it?

Formula	Working	Precise mass
$\text{C}_3\text{H}_4\text{O}_3$	$36.000000 + 4.031300 + 47.984745$	88.0160
$\text{C}_4\text{H}_8\text{O}_2$	$48.000000 + 8.062600 + 31.989830$	88.0524
$\text{C}_3\text{H}_8\text{N}_2\text{O}$	$36.000000 + 8.062600 + 28.006148 + 15.994915$	88.0637
$\text{C}_5\text{H}_{12}\text{O}$	$60.000000 + 12.093900 + 15.994915$	88.0888

Only $\text{C}_4\text{H}_8\text{O}_2$ matches. The nearest rival is 0.011 away, more than a hundred times the precision of the measurement, so the identification is not in doubt. Notice that the molecular formula is all you

have so far: $C_4H_8O_2$ could be ethyl ethanoate, methyl propanoate, propyl methanoate, butanoic acid, 2-methylpropanoic acid, 1,4-dioxane and more besides. Section 7 shows how the other techniques choose between them.

Exam Tip: Carry every decimal place the question gives you all the way to the end and round only at the last step. Neighbouring candidates can be as little as 0.01 apart, so rounding 1.007825 to 1.01 in a molecule with twelve hydrogens moves your answer by 0.026 — further than the gap you are trying to resolve. A quick sanity check that saves time: measure each formula from its *nominal* mass rather than from zero. Every H pushes the precise mass about 0.0078 above the nominal value, every N about 0.0031 above, and every O about 0.0051 below. For $C_4H_8O_2$ that gives $(8 \times 0.0078) - (2 \times 0.0051) \approx +0.052$, which is the +0.0524 observed.

Mass Differences Between Peaks

The second half of this outcome is the one covered in **WM**: the gap between two peaks is the mass of the neutral fragment lost, so a difference of 15 means a CH_3 group has gone, 17 an OH, 18 a water molecule, 29 a CHO or C_2H_5 , 31 a CH_2OH or OCH_3 , 43 a CH_3CO or C_3H_7 , and 45 an OC_2H_5 or $COOH$. Read the gaps *down* from M^+ rather than reading the fragment masses on their own: $M^+ - 15$ tells you the molecule had a methyl group on it, which a bare peak at 15 does not.

High resolution helps here too. A fragment at nominal m/z 43 is CH_3CO^+ at 43.0184 but $C_3H_7^+$ at 43.0548, and a fragment at 29 is CHO^+ at 29.0027 but $C_2H_5^+$ at 29.0391. In each pair, one answer means a carbonyl compound and the other means an alkyl chain.

7. Combining Mass Spectrometry, IR and NMR

No single technique identifies an unknown compound. Mass spectrometry, infrared spectroscopy and NMR each answer a different question, and a structure is only proved when all three sets of data are accounted for by one and the same molecule. This is the skill the H433 papers test most heavily at the top of the mark range, and the marks go to candidates who reason from the evidence rather than recognising the answer.

Technique	What it tells you	What it cannot tell you
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Mass spectrometry	The relative molecular mass from M^+ ; the molecular formula, if the M^+ value is high resolution; which pieces break off, from the gaps between peaks	How the pieces are joined together. Isomers share a molecular formula and often share fragments
Infrared	Which functional groups are present — and, just as usefully, which are absent	How many of each group there are, or where in the molecule they sit
^{13}C NMR	How many different carbon environments there are, and what kind each one is	How many hydrogens are attached (the spectra you are given are proton-decoupled)
^1H NMR	How many different hydrogen environments there are, how many hydrogens in each (integration), what kind each is (shift), and what sits next door (splitting)	Anything about the parts of the molecule with no hydrogens on them

A Method That Works Every Time

1. **Start with the mass spectrum.** Get M_r from M^+ , and the molecular formula if the value is high resolution. Work out how far the formula is from the saturated one: butane is C_4H_{10} , so a $\text{C}_4\text{H}_8\text{O}_2$ molecule is two hydrogens short and must contain one double bond or one ring.
2. **Read the IR for functional groups,** and write down the absences as firmly as the presences. No broad absorption in $2500\text{--}3300\text{ cm}^{-1}$ rules out a carboxylic acid; no broad absorption in $3200\text{--}3600\text{ cm}^{-1}$ rules out an alcohol or phenol.
3. **Count the ^{13}C peaks.** Fewer peaks than carbon atoms means the molecule is symmetrical. Use the shifts to say what kind of carbon each peak is.
4. **Work through the ^1H NMR in the order shift, integration, splitting.** Check the integration adds up to the number of hydrogens in the molecular formula — if it does not, you have misread something.
5. **Assemble the fragments** into a structure, then **test it against every piece of data**, including the data you have not used yet. Finally, name the isomers you have rejected and say what rejected them.

Worked example — determining a structure from combined spectra.

Compound **X** is a colourless liquid. Deduce its structure from the following data.

- **High-resolution mass spectrum:** M^+ at m/z 88.0524. Base peak at m/z 43; further peaks at m/z 45 and m/z 29.
- **Infrared spectrum:** strong sharp absorption at 1742 cm^{-1} ; strong absorptions at 1240 cm^{-1} and 1045 cm^{-1} ; medium absorptions at $2850\text{--}2950\text{ cm}^{-1}$; *nothing* above 3000 cm^{-1} , and nothing broad anywhere.
- ^{13}C NMR (proton-decoupled): four peaks, at δ 171.1, 60.4, 21.0 and 14.2.
- ^1H NMR: δ 4.12 (2H, quartet); δ 2.04 (3H, singlet); δ 1.26 (3H, triplet).

Step 1 — molecular formula. From the worked example in Section 6, m/z 88.0524 is **$\text{C}_4\text{H}_8\text{O}_2$** . The saturated formula for four carbons and two oxygens is $\text{C}_4\text{H}_{10}\text{O}_2$, so X is two hydrogens short: it contains exactly one double bond or one ring.

Step 2 — functional groups from the IR. The sharp 1742 cm^{-1} absorption is a C=O stretch, and it lies in the Data Sheet's **ester** window of $1735\text{--}1750\text{ cm}^{-1}$ rather than the ketone ($1705\text{--}1725$), aldehyde ($1720\text{--}1740$) or carboxylic acid ($1700\text{--}1725$) windows. The strong 1240 and 1045 cm^{-1} absorptions are C–O stretches ($1000\text{--}1300\text{ cm}^{-1}$). That C=O accounts for the one double bond found in Step 1. Now the absences: nothing broad at $2500\text{--}3300\text{ cm}^{-1}$, so *not* a carboxylic acid, which kills butanoic acid and 2-methylpropanoic acid outright; nothing at $3200\text{--}3600\text{ cm}^{-1}$, so no O–H of any kind; nothing at $1620\text{--}1680\text{ cm}^{-1}$, so no C=C. X is an ester, and an ester uses up both oxygen atoms.

Step 3 — carbon environments. Four peaks for four carbon atoms means every carbon is in a different environment, so X has no symmetry. This rules out 1,4-dioxane, the cyclic isomer, which would give a single ^{13}C peak. The shifts confirm the IR: δ 171.1 is a carbonyl carbon ($170\text{--}220$), δ 60.4 is a carbon singly bonded to oxygen ($50\text{--}80$), and δ 21.0 and 14.2 are ordinary alkyl carbons ($5\text{--}40$).

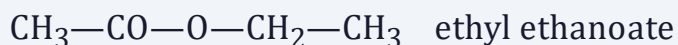
Step 4 — hydrogen environments. The integrations are 2 : 3 : 3, totalling 8H, which matches $\text{C}_4\text{H}_8\text{O}_2$ exactly — no hydrogen is unaccounted for, and no signal has been missed. Taking the three signals in turn:

- δ 1.26, **3H triplet**: a CH_3 in the plain H–C–R alkyl band at the right-hand end of the Data Sheet chart, split into three by *two* neighbouring hydrogens ($n + 1 = 3$, so $n = 2$).

- δ 4.12, **2H quartet**: a CH_2 split into four by *three* neighbouring hydrogens ($n + 1 = 4$, so $n = 3$). These two signals split each other, so the CH_3 and the CH_2 are joined: X contains a CH_3CH_2 -ethyl group.
- δ 2.04, **3H singlet**: a CH_3 with *no* hydrogens on the neighbouring atom. In an ester the only such position is the methyl of a $\text{CH}_3\text{C}=\text{O}$ group, and δ 2.04 sits in the Data Sheet's $\text{H}-\text{C}-\text{C}=\text{O}$ band, around δ 2 to 3, rather than in the plain alkyl band.

The Data Sheet shows the ^1H shifts as a chart of overlapping bands rather than a table, and prints the warning that shifts vary with solvent, concentration and substituents and may fall outside the ranges drawn. Use the bands to decide which *kind* of environment a signal belongs to, not to rule an environment out because a value is a tenth of a ppm outside the bar.

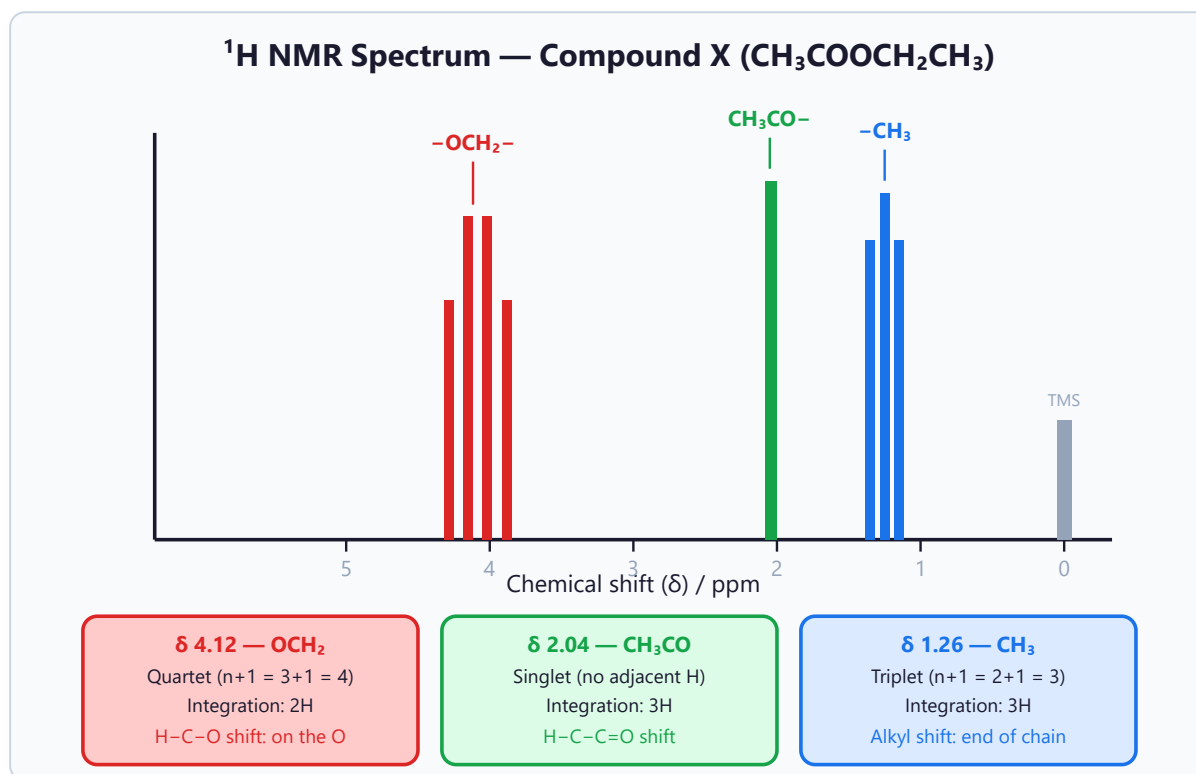
Step 5 — which way round is the ester? This is the step that decides between the two candidates left, ethyl ethanoate and methyl propanoate, and it turns on the *shift* of the quartet. At δ 4.12 the CH_2 is in the Data Sheet's $\text{H}-\text{C}-\text{O}$ band, around δ 3 to 4.5, well downfield of the $\text{H}-\text{C}-\text{C}=\text{O}$ band, because the electronegative oxygen it is attached to withdraws electron density and deshields those protons. So the CH_2 is bonded to the ester oxygen, and the ethyl group is on the **oxygen** side:



Step 6 — test the answer and reject the rivals.

- **Methyl propanoate, $\text{CH}_3\text{CH}_2\text{COOCH}_3$** , gives the same 2H quartet, 3H singlet and 3H triplet, so integration and splitting alone cannot separate it from X. But its quartet is next to the $\text{C}=\text{O}$, at about δ 2.3, and its singlet is the OCH_3 , at about δ 3.7. X has these the other way round. Rejected.
- **Propyl methanoate, $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$** , would show a 1H singlet near δ 8.0 for the $\text{HC}=\text{O}$ hydrogen and a 2H multiplet for the middle CH_2 . Neither is there, and the integration pattern would be 1 : 2 : 2 : 3. Rejected.
- **Butanoic acid and 2-methylpropanoic acid** would show a very broad $\text{O}-\text{H}$ absorption at 2500–3300 cm^{-1} and a 1H signal at δ 10–12. Rejected at Step 2.
- **1,4-dioxane** would give one ^{13}C peak and one ^1H singlet for all 8H, and no $\text{C}=\text{O}$ absorption at all. Rejected at Step 2.

Step 7 — the fragments agree. The base peak at m/z 43 is CH_3CO^+ , the acylium ion left when the molecule breaks at the C–O bond; the complementary fragment is OC_2H_5^+ at m/z 45, and $43 + 45 = 88$. The smaller peak at m/z 29 is C_2H_5^+ . Methyl propanoate would have broken the other way, giving $\text{C}_2\text{H}_5\text{CO}^+$ at m/z 57 and OCH_3^+ at m/z 31, neither of which appears. Every piece of data is now explained by one structure, so **X is ethyl ethanoate**.



Peak heights in the sketch above are drawn for legibility. In a real spectrum it is the *area* under each signal, shown by the integration trace, that is proportional to the number of hydrogens — a tall thin singlet and a short wide quartet can represent the same number of protons.

Exam Tip: In a combined-spectra question the marks are for the reasoning, not for the name at the end. Quote the evidence with the conclusion every time: " 1742 cm^{-1} , in the ester range on the Data Sheet, so C=O in an ester" earns the mark; "it's an ester" does not. Two habits are worth building. First, always say what the *absences* rule out — no broad O–H is usually worth a mark on its own. Second, when two isomers survive everything else, look at the chemical shift of the CH_2 or CH_3 : whichever group sits next to the oxygen is dragged downfield past δ 3.5, and that single number is usually what separates them.

